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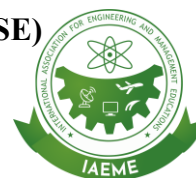
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APPROACH TO EVALUATING AND INTEGRATING OPERATIONAL SAFETY INTO DESIGN

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Abstract

Operational safety is an important consideration in the design of mechanical systems because failures and repair durations directly influence their ability to perform the required function under operating conditions. However, operational-safety assessment is often conducted after equipment commissioning, limiting its usefulness during the design decision process. This study proposes a probabilistic and design-oriented approach for integrating operational safety into mechanical design. Reliability and maintainability are respectively modeled using Weibull and Gumbel distributions, while the parameters of the reliability model are estimated using the maximum likelihood method. A distinct operational-safety formulation is then developed from these probabilistic quantities and used to derive operational-safety, probability-density, risk-rate and confidence-rate functions. The approach is applied to three fire pump units using historical time-between-failure and time-to-repair data. The results demonstrate the applicability of the proposed formulation and show differentiated operational-safety behavior among the studied units. A design-oriented framework is subsequently proposed to use the operational-safety indicator for evaluating, comparing and improving mechanical design solutions. The study provides a quantitative link between reliability, maintainability and operational safety and establishes a basis for incorporating operational-safety considerations earlier in the mechanical design process.

Keywords: Operational Safety; Reliability; Maintainability; Weibull Distribution; Gumbel Distribution; Maximum Likelihood Method; Mechanical Design; Fire Pump.

Abbreviations: TBF: Time Between Failures; TTR: Technical Time to Repair; FPU: Fire Pump Unit; CAD: Computer-Aided Design; MLM: Maximum Likelihood Method; CCPD: Cameroonian Company of Petroleum Depots.

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1. Introduction

Mechanical design is increasingly concerned with the ability of technical systems to perform their intended functions throughout their operating life. The growing complexity of industrial systems, resulting from the integration of mechanical, electrical, hydraulic, control, and information technologies, makes it necessary to consider not only functional and structural requirements but also operational-safety related properties from the early stages of design. In this context, reliability, maintainability, availability, and safety constitute important criteria for assessing the expected performance of a technical system (Avizienis et al. 2000).

Traditionally, mechanical design is structured as a progressive process in which the design problem is defined, alternative solutions are generated, the system architecture is established, and the detailed solution is finally developed. Design methodologies emphasize the iterative nature of this process and the need to validate the proposed solution against predefined requirements (Pahl et al. 2007; Suh et al. 1998). Such an approach provides an appropriate framework for incorporating additional performance criteria during the evaluation and selection of design solutions.

Among these criteria, operational-safety has received considerable attention because the performance of an industrial system depends not only on its ability to perform its intended function but also on its ability to resist failures, to be restored after failure, and to operate without creating unacceptable risks. Reliability characterizes the ability of a system to perform its required function for a specified period, whereas maintainability characterizes the probability of restoring the system to an operational state within a specified time. These two properties are closely related to system availability and operational-safety. Consequently, treating them independently may not fully reflect the combined behavior of an operating system (Avizienis et al. 2000; Nzié et al. 2017).

Several approaches have consequently been developed to evaluate reliability, maintainability, availability, and safety. Classical dependability techniques include Failure Modes, Effects and Criticality Analysis (FMECA), Fault Tree Analysis (FTA), Preliminary Risk Analysis (PRA), and related qualitative or quantitative methods. Probabilistic approaches provide an additional means of representing the variability and uncertainty associated with failure and repair processes. In reliability analysis, statistical distributions such as the Weibull distribution are widely used because their parameters allow different failure-rate behaviors to be represented. Maximum likelihood estimation is also commonly employed for estimating the parameters of reliability models from observed failure data (Perreault and Bobée 1992; Dhillon 2006). Ngnassi Djami et al. (2023) also modeled the failure rate of a standby multi-component system and investigated reliability improvement.

For maintainability analysis, the distribution of repair times is particularly important because the duration of a repair directly influences the time during which a system remains unavailable. Depending on the characteristics and size of the available sample, different statistical distributions can be considered. Extreme-value distributions, including the Gumbel distribution, provide a useful framework for representing repair-time populations characterized by dispersion and extreme observations. The combination of reliability and maintainability models therefore makes it possible to obtain a more comprehensive representation of system operating behavior. Maintenance-service performance can also be investigated using queue-network models (Ngnassi Djami et al., 2024a).

Recent studies have applied probabilistic, statistical, simulation, and optimization techniques to reliability and maintainability assessment in different industrial systems. These include reliability analyses of drilling equipment, load-haul-dump machines, transport systems, production equipment, and other complex systems. Monte Carlo simulation has also been used to evaluate system reliability under different operating conditions (Ngnassi Djami et al., 2024b). Preventive maintenance has further been optimized using exact-resolution methods and genetic algorithms (Ngnassi Djami et al., 2024c). Such contributions demonstrate the importance of probabilistic and optimization-based approaches for dealing with variability in industrial failure and maintenance processes. Ngnassi Djami et al. (2025) also applied a neuro-fuzzy approach to equipment reliability optimization.

However, an important methodological issue remains. In much of the existing work, reliability, maintainability, availability, and safety are evaluated as distinct characteristics, even though these properties interact throughout the operating life of a system. This limitation becomes particularly relevant when the objective is not simply to diagnose the performance of an existing system but to use operational-safety information to support the selection and validation of a design solution.

The integration of operational-safety into design has therefore become an important research topic. Cressent et al. (2011) proposed an approach for integrating operational-safety analyses into the system design process. Coulibaly et al. (2008) investigated the consideration of maintainability and safety indicators during the design of mechanical products, while Menye (2009) addressed the validation of maintainability and availability in multicomponent systems. Nabdi and Herrou (2015) considered the integration of maintenance requirements into design. More recent contributions have investigated the consideration of operational safety in complex-system architecture, integrated reliability analysis and design, and the incorporation of reliability, availability, and maintainability into model-based systems engineering (Baklouti 2020; Pahl 2021; Diatte et al. 2022). Reliability-oriented design approaches have also been developed for structural and automotive systems (Rajan et al. 2020; Piancastelli et al. 2026; Ngnassi Djami et al., 2020a; Ngnassi Djami et al., 2020b; Ngnassi Djami et al., 2020c).

These studies demonstrate that reliability and maintainability can effectively contribute to design decisions. Nevertheless, the integration of a combined operational-safety indicator based simultaneously on reliability, maintainability, and safety remains insufficiently addressed. In particular, there is a need for a quantitative procedure capable of linking probabilistic operational-safety assessment to the evaluation and validation of a candidate mechanical design.

This issue motivated the authors' previous work, in which a probabilistic model was proposed to evaluate operational safety by combining reliability, maintainability, and safety and was applied to equipment belonging to the Cameroonian Company of Petroleum Depots (CCPD) in Garoua (Tatsinkou et al. 2024). That study established the basis for a probabilistic representation of operational safety and demonstrated its application to industrial equipment. The present work constitutes a continuation of that research, but it pursues a different objective and uses a different formulation of the operational-safety model.

More specifically, the present study does not simply reproduce the operational-safety formulation developed previously. The probabilistic methodology used to construct the model and the general principles of reliability and maintainability estimation are retained, but the mathematical formulation defining operational safety is modified. This change leads to a different sequence of transformations for the operational-safety function and its associated probability density, risk-rate, and confidence-rate functions. The distinction between the two formulations is therefore fundamental to the scientific contribution of the present work.

The objective of this study is consequently twofold: (i) to develop and evaluate the proposed probabilistic formulation of operational safety using reliability and maintainability models estimated from industrial data, and (ii) to demonstrate how the resulting operational-safety indicator can be incorporated into the mechanical design process as a criterion for evaluating and validating candidate solutions.

To achieve this objective, the study uses the Weibull distribution to model reliability and the Gumbel distribution to represent maintainability. The Maximum Likelihood Method (MLM) is employed for the estimation of the reliability parameters, while the maintainability parameters are obtained from the analysis of repair-time data. The Kolmogorov–Smirnov test is used to assess the adequacy of the Weibull model for the observed TBF data. The proposed approach is then applied to the fire pump units of the CCPD facility in Garoua. The resulting operational-safety functions are analyzed together with their probability density, risk-rate, and confidence-rate characteristics (Tatsinkou et al. 2024).

Finally, based on these results, a design-oriented metamodel is proposed to integrate operational-safety into the mechanical design process. The metamodel establishes a feedback mechanism linking design requirements, CAD information, operational data, probabilistic modeling, operational-safety evaluation, and design validation. A candidate solution is considered eligible when its estimated system-level operational safety satisfies the level specified in the design requirements; otherwise, the design is reconsidered and re-evaluated. This type of maintenance-asset metamodeling is also consistent with the approach reported by Ngnassi Djami et al. (2025).

The remainder of this paper is organized as follows. Section 2 presents the probabilistic formulation used to evaluate operational safety. Section 3 describes the Maximum Likelihood Method and its application to the proposed model. Section 4 presents and discusses the results obtained from the CCPD fire pump units. Section 5 develops the proposed approach for integrating operational safety into mechanical design. Section 6 summarizes the main conclusions and identifies directions for future research.

2. Methods

2.1. Probabilistic representation of reliability, maintainability and safety

The proposed approach considers operational safety as a time-dependent property resulting from the interaction between the operating behaviour of a system and its repair behaviour. Two random variables are therefore considered: the time Between Failures, TBF, and the technical time to repair, TTR. Their realizations are denoted by t_f and t_r , respectively.

The reliability of the system at operating time t_f is defined as the probability that the system remains operational beyond t_f . It is expressed as

$$R(t) = P(T_f > t_f) = 1 - F(t_f) \quad (1)$$

where $F(t_f)$ is the cumulative distribution function of the failure time.

For a continuous random variable, the cumulative distribution function is obtained from its probability density function $f(t)$ according to

$$F(t) = \int_0^t f(\tau) d\tau \quad (2)$$

The instantaneous failure rate is defined as the ratio between the probability density of failure and the corresponding reliability:

$$\lambda(t) = \frac{f(t)}{R(t)} \quad (3)$$

Consequently, the reliability function can also be represented in terms of the cumulative failure rate:

$$R(t_f) = e^{-\int_0^{t_f} \lambda(\tau) d\tau} \quad (4)$$

The maintainability function describes the probability that a failed system is restored within a specified repair time. It is therefore defined from the repair-time random variable as:

$$M(t_r) = P(T_r < t_r) = \int_0^{t_r} g(\tau) d\tau \quad (5)$$

where $g(t)$ denotes the probability density function of the technical time to repair.

Using the repair-rate function $\mu(t)$, maintainability can equivalently be written as:

$$M(t_r) = 1 - e^{-\int_0^{t_r} \mu(\tau) d\tau} \quad (6)$$

The repair rate is defined analogously to the failure rate:

$$\mu(t) = \frac{g(t)}{1-M(t)} \quad (7)$$

In this study, safety is considered in relation to the measures implemented to maintain equipment in an acceptable operating condition and to limit hazardous consequences during operation and maintenance. In particular, the formulation adopted here follows the interpretation that operational safety depends on the interaction between reliability and maintainability. This conceptual relationship is consistent with the safety-oriented framework previously discussed by Nzié et al. (2017).

2.2. Proposed probabilistic formulation of operational safety

The fundamental quantities defined above provide the probabilistic variables required to formulate operational safety. In the present study, operational safety is not obtained by reproducing the formulation introduced in the authors' previous work. Instead, a different probabilistic composition of reliability and maintainability is introduced.

Let

$$A = \{T_f > t_f\}$$

denote the event that the system remains operational during the considered operating period, and let

$$B = T_r < t_r$$

denote the event that the system can be restored within the considered repair period. Hence,

$$P(A) = R(t_f)$$

and

$$P(B) = M(t_r)$$

The present model combines the joint occurrence of these two conditions with their union. The proposed operational-safety index is consequently formulated as

$$OpS(t_f, t_r) = P \left[(T_f > t_f) \cap (T_r < t_r) \cap [(T_f > t_f) \cup (T_r < t_r)] \right] \quad (9)$$

Under the probabilistic representation adopted in this study, the two terms are expressed as

$$P(A \cap B) = R(t_f)M(t_r)$$

and

$$P(A \cup B) = R(t_f) + M(t_r) - R(t_f)M(t_r)$$

The operational-safety function consequently becomes

$$OpS(t_f, t_r) = R(t_f)M(t_r)[R(t_f) + M(t_r) - R(t_f)M(t_r)] \quad (10)$$

This formulation constitutes the principal mathematical distinction between the present model and the formulation developed in the authors' previous study. In the latter, operational safety was represented by a complementary expression involving the squared product of the reliability and maintainability deficits. The present formulation instead introduces a direct multiplicative coupling between the joint reliability–maintainability term and their union probability. The two formulations therefore generate different operational-safety surfaces even when identical reliability and maintainability functions are considered. The previous formulation is reported in the published study of Tatsinkou et al. (2024).

This distinction is important because the change introduced in Eq. (10) is propagated through all subsequent quantities derived from the operational-safety function. Thus, the probability density, risk rate, confidence rate and likelihood function used for parameter estimation are specific to the present formulation.

2.3. Operational unsafety, probability density, risk and confidence

The complementary measure of operational safety is defined as the operational-unsafety function:

$$OpUS(t_f, t_r) = 1 - OpS(t_f, t_r) \quad (11)$$

The operational-safety formulation defined in Eq. (10) provides the basis for deriving the probability density associated with the two-dimensional operating/repair-time representation. In accordance with the formulation adopted in this study, the probability density is obtained from the corresponding derivatives of the operational-safety function. The resulting expression is given in Eq. (12), while its expanded form is given in Eq. (13).

$$l(t_f, t_r) = \frac{\partial^2 OpUS(t_f, t_r)}{\partial t_f \partial t_r} = -\frac{\partial R(t_f)^2}{\partial t_f} \frac{\partial M(t_r)}{\partial t_r} - \frac{\partial R(t_f)}{\partial t_f} \frac{\partial M(t_r)^2}{\partial t_r} + \frac{\partial R(t_f)^2}{\partial t_f} \frac{\partial M(t_r)}{\partial t_r} \quad (12)$$

$$= 2\lambda(t)\mu(t)R(t_f)^2[1 - M(t_r)] + 2\lambda(t)\mu(t)R(t_f)[M(t_r) - M(t_r)^2] - 4\lambda(t)\mu(t)R(t_f)^2[M(t_r) - M(t_r)^2] \quad (13)$$

The probability density obtained from the proposed operational-safety formulation provides the basis for defining the risk rate. The latter is expressed as the ratio between the probability density and the operational-safety function:

$$\varphi(t_f, t_r) = \frac{l(t_f, t_r)}{OpS(t_f, t_r)} \quad (14)$$

where $l(t_f, t_r)$ denotes the probability-density function associated with the proposed operational-safety formulation.

The corresponding confidence rate is defined as the complement of the risk rate:

$$\psi(t_f, t_r) = 1 - \varphi(t_f, t_r) \quad (15)$$

Equations (12) – (15) are retained from the mathematical development of the present manuscript and are subsequently used in the numerical evaluation. In particular, Eqs. (14) and (15) do not require an independent parameter-estimation procedure because they are directly determined from the operational-safety function and its associated probability density. The manuscript likewise indicates that the risk and confidence rates are derived from the operational-safety and density functions.

2.4. Distinction between the present formulation and the previous model

For clarity, the relationship between the two studies can be summarized as follows. The previous model proposed by Tatsinkou et al. (2024) was based on the formulation

$$OpSP(t_f, t_r) = 1 - [(R(t_f) - 1)(1 - M(t_r))]^2$$

Its subsequent probability-density, risk-rate and confidence-rate expressions were derived from this formulation.

The present study adopts instead

$$OpSN(t_f, t_r) = R(t_f)M(t_r)[R(t_f) + M(t_r) - R(t_f)M(t_r)]$$

Consequently,

$$OpS_P \neq OpS_N$$

and, because the subsequent quantities are derived from their respective operational-safety functions,

$$l_N \neq l_P, \quad \varphi_N \neq \varphi_P, \quad \psi_N \neq \psi_P$$

This distinction is not related to a change in the general estimation strategy. Rather, it originates from the different probabilistic formulation of operational safety. The same general modelling procedure — definition of reliability and maintainability, selection of appropriate probability laws and estimation of their parameters — can therefore be retained while producing a different operational-safety model.

This point is particularly important for the subsequent Maximum Likelihood Method (MLM), because the likelihood function in the present study is constructed from the probability density associated with the new operational-safety formulation rather than from the density developed in the previous study.

2.5. Methodological pathway

The complete methodological sequence used in this study can therefore be represented as

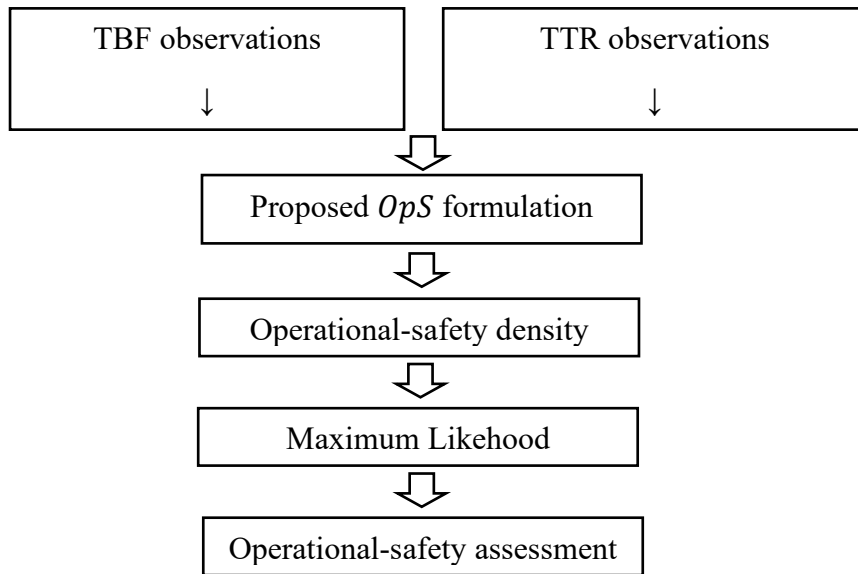
$$(TBF, TTR) \rightarrow R(t_f), M(t_r) \rightarrow OpS(t_f, t_r) \rightarrow l(t_f, t_r) \rightarrow \varphi(t_f, t_r), \psi(t_f, t_r)$$

followed by the statistical characterization of the two fundamental random variables.

For the operating time, a two-parameter Weibull distribution is adopted. Its shape parameter enables the model to represent different failure-rate behaviours, including increasing failure rates associated with ageing. For the repair time, a Gumbel distribution is retained because of its suitability for asymmetric repair-time observations and extreme repair durations. The current manuscript explicitly specifies the use of Weibull for reliability and Gumbel for maintainability.

The resulting functions are subsequently introduced into the new operational-safety formulation to obtain the Weibull–Gumbel representation used for parameter estimation. The manuscript then applies the MLM to the resulting operational-safety probability density and derives the corresponding likelihood equations.

The methodological structure is therefore:



This sequence provides the mathematical basis for the application to the fire pump units and, subsequently, for the integration of operational-safety assessment into the mechanical design process.

3. Parameter Estimation and Operational-Safety Assessment

3.1. Maximum likelihood formulation

The maximum likelihood method (MLM) provides a statistical framework for estimating the parameters of a probabilistic model from observed failure-time data. Let t_i , $i = 1, \dots, n$, denote the observed failure times and let η represent the vector of unknown model parameters. The likelihood function corresponds to the probability of observing the available data for a given parameter vector.

For the reliability analysis considered in this study, the likelihood function is expressed by Eq. (16). The maximum likelihood estimators are obtained by determining the parameter values that maximize this function. Equivalently, when the likelihood function is differentiable, the estimators satisfy the first-order conditions given by Eq. (17). For numerical convenience, the corresponding logarithmic likelihood formulation is given by Eq. (18). These three expressions establish the general estimation framework used subsequently for the Weibull operational-safety model.

$$L(\eta | t_1, t_2, \dots, t_n) = \prod_{i=1}^n f(t_i; \eta) \quad (16)$$

where $f(t_i; \eta)$ denotes the probability density associated with the observation t_i .

$$\left. \frac{\partial L(t_1, t_2, \dots, t_n; \eta)}{\partial \eta_j} \right|_{\eta = \hat{\eta}} = 0, \quad j = 1, \dots, p \quad (17)$$

$$\left. \frac{\partial \ln(t_1, t_2, \dots, t_n; \eta)}{\partial \eta_j} \right|_{\eta = \hat{\eta}} = 0, \quad j = 1, \dots, p \quad (18)$$

3.2. Reliability and maintainability probability laws

The reliability model adopted in this study is based on the two-parameter Weibull distribution (Ngnassi Djami et al., 2024d). This distribution is particularly suitable for mechanical equipment because its shape parameter allows different failure-rate behaviors to be represented. The reliability function is therefore expressed according to Eq. (19), where θ denotes the shape parameter and α the scale parameter.

$$R(t) = e^{-\left(\frac{t}{\alpha}\right)^\theta} \quad (19)$$

The maintainability behavior is represented by a Gumbel distribution. In this formulation, the parameters a and u characterize, respectively, the slope-related parameter and the location parameter of the Gumbel representation. The maintainability function is given by Eq. (20).

$$M(t) = e^{-e^{-a(t-u)}} \quad (20)$$

3.3 Weibull–Gumbel formulation of operational safety

Substituting the Weibull reliability function and the Gumbel maintainability function into the operational-safety formulation developed in this study leads to the analytical expression given by Eq. (21). This equation constitutes the central probabilistic relationship used for the subsequent evaluation of operational safety.

$$\begin{aligned} OpS(t_f, t_r) = & \left[e^{-2\left(\frac{t_f}{\alpha_f}\right)^{\theta_f}} \right] \left[e^{-e^{-a(t_r-u)}} \right] + \left[e^{-\left(\frac{t_f}{\alpha_f}\right)^{\theta_f}} \right] \left[e^{-e^{-a(t_r-u)}} \right]^2 \\ & - \left[e^{-2\left(\frac{t_f}{\alpha_f}\right)^{\theta_f}} \right] \left[e^{-e^{-a(t_r-u)}} \right]^2 \end{aligned} \quad (21)$$

The formulation in Eq. (21) is specific to the operational-safety model proposed in the present study. In particular, it results from the probabilistic structure established in Eq. (9) and its analytical development in Eqs. (10) – (15). Consequently, the subsequent Weibull–Gumbel formulation should not be interpreted as a direct repetition of the model developed in the authors' previous study. Although the same fundamental reliability and maintainability concepts are used, the operational-safety formulation is different, leading to a different analytical development.

The resulting expression provides the basis for calculating the operational-safety function as a function of failure time and repair time. It also permits the derivation of the corresponding probability density and other operational-safety indicators.

3.4 Probability density and likelihood of the operational-safety model

The estimation procedure requires a probability density function associated with the operational-safety model. Starting from the operational-safety expression in Eq. (21), the corresponding density is obtained by differentiation, taking into account the Weibull failure-rate and Gumbel repair-rate functions introduced previously.

The resulting density function is given by Eq. (22).

$$l(t_f, t_r) = - \left[2 \left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_f}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_f}{\alpha_f} \right)^{\theta_f}} \right) \right] \left[a e^{-a(t_r-u)} e^{-e^{-a(t_r-u)}} \right] -$$

$$\left[\left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_f}{\alpha_f} \right)^{\theta_f-1} \left(e^{-\left(\frac{t_f}{\alpha_f} \right)^{\theta_f}} \right) \right] \left[2 a e^{-a(t_r-u)} e^{-2 e^{-a(t_r-u)}} \right] +$$

$$\left[2 \left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_f}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_f}{\alpha_f} \right)^{\theta_f}} \right) \right] \left[2 a e^{-a(t_r-u)} e^{-2 e^{-a(t_r-u)}} \right]$$
(22)

The density function defined by Eq. (22) is then introduced into the general likelihood formulation of Eq. (16). For n independent observations, this leads to the likelihood expression given in Eq. (23).

$$L(t_1, t_2, \dots, t_n; \theta) = \prod_{i=1}^n f(t_i, \theta, \alpha) = \prod_{i=1}^n l(t_{fi}, t_{ri}, \theta_f, \alpha_f)$$

$$= \prod_{i=1}^n \left[-2 \left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \right] \prod_{i=1}^n \left[a e^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right] +$$

$$\prod_{i=1}^n \left[- \left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \right] \prod_{i=1}^n \left[2 a e^{-a(t_{ri}-u)} e^{-2 e^{-a(t_{ri}-u)}} \right] +$$

$$\prod_{i=1}^n \left[2 \left(\frac{\theta_f}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \right] \prod_{i=1}^n \left[2 a e^{-a(t_{ri}-u)} e^{-2 e^{-a(t_{ri}-u)}} \right]$$
(23)

The estimation of the Weibull parameters is subsequently performed by maximizing this likelihood function. Differentiating Eq. (23) with respect to the scale parameter α gives Eq. (24), whereas differentiation with respect to the shape parameter θ gives Eq. (25).

$$\frac{\partial L}{\partial \alpha_f} = \left[-2 a e^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - 2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right] -$$

$$\left[2 a e^{-a(t_{ri}-u)} e^{-2 e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right] +$$

$$\left[4 a e^{-a(t_{ri}-u)} e^{-2 e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - 2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right]$$
(24)

$$\frac{\partial L}{\partial \theta_f} = \left[-2 a e^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right]^n \times$$

$$\sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - 2 \theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right] -$$

$$\begin{aligned}
& \left[2ae^{-a(t_{ri}-u)} e^{-2e^{-a(t_{ri}-u)}} \right]^n \times \\
& \sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - \theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right] + \\
& \left[2a \left(e^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right) \right]^n \times \\
& \sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - 2\theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right]
\end{aligned} \tag{25}$$

The maximum likelihood estimates are therefore obtained by solving the system resulting from the first-order conditions. This system is expressed by Eq. (26).

$$\begin{cases} \frac{\partial \ln L}{\partial \alpha_f} = 0 \\ \frac{\partial \ln L}{\partial \theta_f} = 0 \end{cases} \tag{26}$$

Because the resulting equations are nonlinear, an analytical closed-form solution is not directly available. The parameter estimates are consequently obtained by solving the nonlinear system numerically.

3.5 Successive approximation and initialization of the Weibull parameters

The nonlinear estimation problem defined by Eqs. (26) – (28) is solved using a successive-approximation procedure. The procedure requires suitable initial values for the Weibull parameters before numerical optimization is performed.

The equations governing the estimation system are given by Eqs. (27) and (28).

$$\begin{aligned}
& \left[-2ae^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - 2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right] - \\
& \left[2ae^{-a(t_{ri}-u)} e^{-2e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right] + \\
& \left[4ae^{-a(t_{ri}-u)} e^{-2e^{-a(t_{ri}-u)}} \right]^n \sum_{i=1}^n \left[\left(\frac{\theta_f}{\alpha_f} \right)^2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 - 2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \right] = 0
\end{aligned} \tag{27}$$

$$\begin{aligned}
& \left[-2ae^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right]^n \times \\
& \sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f-1} \left(e^{-2\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - 2\theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right] - \\
& \left[2ae^{-a(t_{ri}-u)} e^{-2e^{-a(t_{ri}-u)}} \right]^n \times
\end{aligned} \tag{28}$$

$$\sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f - 1} \left(e^{-\left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - \theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right] +$$

$$\left[2a \left(e^{-a(t_{ri}-u)} e^{-e^{-a(t_{ri}-u)}} \right) \right]^n \times$$

$$\sum_{i=1}^n \left[\left(\frac{1}{\alpha_f} \right) \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f - 1} \left(e^{-2 \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f}} \right) \left(1 + \theta_f \ln \left(\frac{t_{fi}}{\alpha_f} \right) - 2\theta_f \left(\frac{t_{fi}}{\alpha_f} \right)^{\theta_f} \right) \ln \left(\frac{t_{fi}}{\alpha_f} \right) \right] = 0$$

An initial estimate of the Weibull shape parameter is obtained from the relationship between the experimental mean of the observed TBF sample and its theoretical Weibull mean. This relationship involves the gamma function and is expressed by Eq. (29).

$$\frac{\sum t_i}{n} = \left(\frac{\sum t_i}{n} \right)^{\frac{1}{\theta}} \times \Gamma \left(1 + \frac{1}{\theta} \right) \quad (29)$$

For each fire pump unit (FPU), the observed TBF values are then introduced into this relationship to determine an appropriate starting value for θ . The calculation illustrated for FPU No. 1 is given by Eq. (30).

$$\frac{\sum_{i=1}^n TBF_i}{n} = 35.45 \quad (30)$$

The subsequent relationships, given by Eqs. (31) – (33), complete the calculation of the initial parameter values.

$$\frac{\sum_{i=1}^n TBF_i^{\theta}}{n} = 36.702 \quad (31)$$

$$\Gamma \left(1 + \frac{1}{\theta} \right) = 0.966 \quad (32)$$

and

$$\left(\frac{\sum_{i=1}^n TBF_i^{\theta}}{n} \right) \times \Gamma \left(1 + \frac{1}{\theta} \right) = 35.4694 \quad (33)$$

The same procedure is applied to the other fire pump units. The initial value of the scale parameter is obtained from the sample mean, whereas the shape parameter is determined from the theoretical relationship involving the gamma function. The resulting initial estimates are then used as starting points for the successive-approximation procedure.

$$OpS_1(t_f, t_r) = \left[e^{-2 \left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[e^{-e^{-0.683(t_r-1)}} \right] + \left[e^{-\left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[e^{-2e^{-0.683(t_r-1)}} \right]$$

$$- \left[e^{-2 \left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[e^{-2e^{-0.683(t_r-1)}} \right] \quad (34)$$

$$l_1(t_f, t_r) = \left[0.067 \left(\frac{t_f}{35.45} \right)^{0.2} e^{-2 \left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[0.683 e^{-0.683(t_r-1)} e^{-e^{-0.683(t_r-1)}} \right]$$

$$\begin{aligned}
 & + \left[0.033 \left(\frac{t_f}{35.45} \right)^{0.2} e^{-\left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[1.336 e^{-0.683(t_r-1)} e^{-2e^{-0.683(t_r-1)}} \right] \\
 & - \left[0.067 \left(\frac{t_f}{35.45} \right)^{0.2} e^{-\left(\frac{t_f}{35.45} \right)^{1.2}} \right] \left[1.336 e^{-0.683(t_r-1)} e^{-2e^{-0.683(t_r-1)}} \right]
 \end{aligned} \tag{35}$$

$$\begin{aligned}
 OpS_2(t_f, t_r) &= \left[e^{-2\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[e^{-e^{-0.725(t_r-0.825)}} \right] + \left[e^{-\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[e^{-2e^{-0.725(t_r-0.825)}} \right] \\
 & - \left[e^{-2\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[e^{-2e^{-0.725(t_r-0.825)}} \right]
 \end{aligned} \tag{36}$$

$$\begin{aligned}
 l_2(t_f, t_r) &= \left[0.048 \left(\frac{t_f}{62.278} \right)^{0.5} e^{-2\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[0.725 e^{-0.725(t_r-0.825)} e^{-e^{-0.725(t_r-0.825)}} \right] \\
 & + \left[0.024 \left(\frac{t_f}{62.278} \right)^{0.5} e^{-\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[1.456 e^{-0.725(t_r-0.825)} e^{-2e^{-0.725(t_r-0.825)}} \right] \\
 & - \left[0.048 \left(\frac{t_f}{62.278} \right)^{0.5} e^{-2\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[1.456 e^{-0.725(t_r-0.825)} e^{-2e^{-0.725(t_r-0.825)}} \right]
 \end{aligned} \tag{37}$$

$$\begin{aligned}
 OpS_3(t_f, t_r) &= \left[e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-e^{-0.793(t_r-1.07)}} \right] + \left[e^{-\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-2e^{-0.793(t_r-1.07)}} \right] \\
 & - \left[e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-2e^{-0.793(t_r-1.07)}} \right]
 \end{aligned} \tag{38}$$

$$\begin{aligned}
 l_3(t_f, t_r) &= \left[0.067 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[0.793 e^{-0.793(t_r-1.07)} e^{-e^{-0.793(t_r-1.07)}} \right] \\
 & + \left[0.025 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[1.586 e^{-0.793(t_r-1.07)} e^{-2e^{-0.793(t_r-1.07)}} \right] \\
 & - \left[0.067 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[1.586 e^{-0.793(t_r-1.07)} e^{-2e^{-0.793(t_r-1.07)}} \right]
 \end{aligned} \tag{39}$$

The numerical solution is obtained using the successive-approximation procedure implemented in the statistical computing environment used in the study (Logiciel R). The resulting optimal values of α and θ , together with the Gumbel parameters a and u , are subsequently reported and analyzed in Section 4.

3.6 Operational-safety evaluation procedure

Once the Weibull and Gumbel parameters have been estimated, the operational-safety function can be evaluated for any specified combination of TBF and TTR values. The procedure consists of the following sequence:

1. Collection of TBF and TTR observations for the equipment under study ;
2. Selection of appropriate probability laws for failure and repair times ;
3. Verification of the adequacy of the selected reliability model using a goodness-of-fit procedure ;
4. Estimation of the Weibull parameters using the maximum likelihood approach;
5. Estimation of the Gumbel parameters from the repair-time representation;
6. Substitution of the estimated parameters into the operational-safety formulation ;
7. Calculation of operational-safety, probability-density, risk-rate and confidence-rate functions ;
8. Graphical representation of these functions over the relevant TBF and TTR domains.

The numerical results obtained from this procedure, including the goodness-of-fit analysis, the estimated parameters and the operational-safety curves, are presented in Section 4.

4. Results and Discussion: Application to the CCPD Equipment in Garoua

4.1. Case-study equipment and operational data

The proposed operational-safety model was applied to three fire pump units (FPUs) operated by the Cameroonian Company of Petroleum Depots (CCPD) in Garoua. These units were selected because they are locally monitored and represent critical equipment within the studied installation. In addition, previous failure analyses, including the FMECA reported by Tatsinkou et al. (2024), identified the fire pump units among the equipment requiring particular attention.

Figures 3 and 4 illustrate the fire pump units considered in the application.



Fig. 3 View of fire pump unit No. 1 and 3



Fig. 4 View of fire pump unit No. 2

The analysis uses historical operating data collected over a two-year period. For each FPU, 14 observations of time between failures (TBF) and 14 observations of technical time to repair (TTR) are available. These data constitute the operational basis for estimating the reliability and maintainability models and for subsequently evaluating the proposed operational-safety functions.

The historical TBF and TTR observations are summarized in Table 1.

Table 1 Historical TBF and TTR data for FPU Nos. 1, 2 and 3 (in hours)

TBF FPU No. 1	TTR FPU No. 1	TBF FPU No. 2	TTR FPU No. 2	TBF FPU No. 3	TTR FPU No. 3
43.50	2.10	27.30	1.40	19.50	1.20
27.20	1.15	39.70	1.15	31.20	0.50
98.30	2.50	18.20	2.00	78.30	3.10
21.10	0.80	57.70	0.50	51.10	1.25
14.10	5.50	96.30	0.60	94.10	1.00
19.40	1.75	88.40	1.00	19.40	1.10
28.80	0.50	123.90	0.50	26.80	2.00
25.90	0.50	75.30	1.50	103.10	1.25
21.40	1.50	83.80	7.00	8.40	1.00
12.70	1.20	64.70	0.90	48.10	0.70
29.30	0.60	71.70	1.50	72.90	1.20
56.70	0.75	25.70	0.50	36.20	3.60
33.50	1.00	42.80	0.70	17.90	1.40
64.40	2.00	56.40	1.40	45.60	2.40

The use of these historical observations is particularly appropriate for the present case study because the objective is not to introduce a new operational database, but to assess the behavior of the proposed operational-safety formulation using an existing industrial dataset.

4.2. Goodness-of-fit analysis

Before estimating the reliability parameters, the adequacy of the Weibull distribution was examined using the Kolmogorov–Smirnov (K–S) goodness-of-fit test at a 5% significance level.

Table 2 presents the detailed K–S calculations for FPU No. 1. The observed TBF values are first arranged in ascending order and then compared with the corresponding theoretical Weibull probabilities. The largest absolute difference between the empirical and theoretical distributions is used as the K–S statistic.

Table 2 Kolmogorov-Smirnov test for FPU No. 1

i	TBF ordered	$F(t) = \frac{i - 0.3}{N - 0.4}$	$R(t) = e^{-\left(\frac{t}{\alpha}\right)^\theta}$	$F_r(t) = 1 - R(t)$	$ F_r(t) - F_{th}(t) $
1	12.70	0.048	0.941	0.058	0.010
2	14.10	0.118	0.921	0.078	0.039
3	19.40	0.187	0.811	0.188	0.001
4	21.10	0.256	0.765	0.234	0.022
5	21.40	0.326	0.756	0.243	0.083
6	25.90	0.395	0.613	0.386	0.009
7	27.30	0.465	0.565	0.434	0.031
8	28.80	0.534	0.513	0.486	0.048
9	29.30	0.604	0.493	0.503	0.100
10	33.50	0.673	0.354	0.645	0.028
11	43.50	0.743	0.107	0.892	0.149
12	56.70	0.812	0.007	0.992	0.179
13	64.40	0.881	0.000	0.999	0.117
14	98.30	0.951	0.000	1.000	0.048

For FPU No. 1, the calculated K–S statistic is lower than the corresponding critical value at the 5% significance level. Consequently, the null hypothesis that the observed TBF data are consistent with a Weibull distribution is not rejected at the 5% significance level.

The same procedure was applied to FPU Nos. 2 and 3. The resulting statistics likewise support the use of the Weibull distribution for representing the TBF data of the three fire pump units. This result provides the statistical basis for using the Weibull model in the subsequent operational-safety calculations.

4.3. Estimated model parameters

Following the goodness-of-fit analysis, the Weibull parameters were estimated using the maximum likelihood procedure described in Section 3. The Gumbel parameters associated with the repair-time distribution were obtained from the corresponding Gumbel representation.

The resulting parameter estimates are presented in Table 3.

Table 3 Optimal values of the different FPU parameters

parameters	α	θ	a	u
FPU No. 1	35.340	1.200	0.182	1.000
FPU No. 2	62.278	1.500	0.725	0.825
FPU No. 3	46.614	1.200	0.793	1.070

The estimated Weibull shape parameters are greater than unity for all three FPU. Within the Weibull framework, this indicates an increasing failure rate with operating time and is therefore consistent with a wear-out tendency. This result is important from a maintenance perspective because equipment operating in such a regime requires particular attention to degradation monitoring and maintenance planning.

The estimated parameters are of the same order as those obtained in the previous study using the same historical TBF dataset. This consistency is expected because the underlying failure observations are unchanged. However, the present parameter estimates are subsequently introduced into the distinct operational-safety formulation developed in this study.

For the three FPUs, the estimated parameters lead directly to the operational-safety and probability-density expressions given by Eqs. (40) – (45).

4.4. Operational-safety and probability-density functions

For FPU No. 1, substitution of the estimated parameters into the operational-safety model gives the expression in Eq. (40), while the corresponding probability-density function is given by Eq. (41).

$$\begin{aligned} OpS_1(t_f, t_r) = & \left[e^{-2\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[e^{-e^{-0.683(t_r-1)}} \right] + \left[e^{-\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[e^{-2e^{-0.683(t_r-1)}} \right] \\ & - \left[e^{-2\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[e^{-2e^{-0.683(t_r-1)}} \right] \end{aligned} \quad (40)$$

$$\begin{aligned} l_1(t_f, t_r) = & \left[0.067 \left(\frac{t_f}{35.45}\right)^{0.2} e^{-2\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[0.683 e^{-0.683(t_r-1)} e^{-e^{-0.683(t_r-1)}} \right] \\ & + \left[0.033 \left(\frac{t_f}{35.45}\right)^{0.2} e^{-\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[1.336 e^{-0.683(t_r-1)} e^{-2e^{-0.683(t_r-1)}} \right] \\ & - \left[0.067 \left(\frac{t_f}{35.45}\right)^{0.2} e^{-2\left(\frac{t_f}{35.45}\right)^{1.2}} \right] \left[1.336 e^{-0.683(t_r-1)} e^{-2e^{-0.683(t_r-1)}} \right] \end{aligned} \quad (41)$$

For FPU No. 2, the corresponding operational-safety and probability-density functions are expressed by Eqs. (42) and (43), respectively.

$$\begin{aligned} OpS_2(t_f, t_r) = & \left[e^{-2\left(\frac{t_f}{62.278}\right)^{1.5}} \right] \left[e^{-e^{-0.725(t_r-0.825)}} \right] + \left[e^{-\left(\frac{t_f}{62.278}\right)^{1.5}} \right] \left[e^{-2e^{-0.725(t_r-0.825)}} \right] \\ & - \left[e^{-2\left(\frac{t_f}{62.278}\right)^{1.5}} \right] \left[e^{-2e^{-0.725(t_r-0.825)}} \right] \end{aligned} \quad (42)$$

$$l_2(t_f, t_r) = \left[0.048 \left(\frac{t_f}{62.278}\right)^{0.5} e^{-2\left(\frac{t_f}{62.278}\right)^{1.5}} \right] \left[0.725 e^{-0.725(t_r-0.825)} e^{-e^{-0.725(t_r-0.825)}} \right]$$

$$\begin{aligned}
 & + \left[0.024 \left(\frac{t_f}{62.278} \right)^{0.5} e^{-\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[1.456 e^{-0.725(t_r-0.825)} e^{-2e^{-0.725(t_r-0.825)}} \right] \\
 & - \left[0.048 \left(\frac{t_f}{62.278} \right)^{0.5} e^{-2\left(\frac{t_f}{62.278} \right)^{1.5}} \right] \left[1.456 e^{-0.725(t_r-0.825)} e^{-2e^{-0.725(t_r-0.825)}} \right]
 \end{aligned}
 \tag{43}$$

Similarly, for FPU No. 3, the corresponding expressions are given by Eqs. (44) and (45).

$$\begin{aligned}
 OpS_3(t_f, t_r) = & \left[e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-e^{-0.793(t_r-1.07)}} \right] + \left[e^{-\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-2e^{-0.793(t_r-1.07)}} \right] \\
 & - \left[e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[e^{-2e^{-0.793(t_r-1.07)}} \right]
 \end{aligned}
 \tag{44}$$

$$\begin{aligned}
 l_3(t_f, t_r) = & \left[0.067 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[0.793 e^{-0.793(t_r-1.07)} e^{-e^{-0.793(t_r-1.07)}} \right] \\
 & + \left[0.025 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[1.586 e^{-0.793(t_r-1.07)} e^{-2e^{-0.793(t_r-1.07)}} \right] \\
 & - \left[0.067 \left(\frac{t_f}{46.614} \right)^{0.2} e^{-2\left(\frac{t_f}{46.614} \right)^{1.2}} \right] \left[1.586 e^{-0.793(t_r-1.07)} e^{-2e^{-0.793(t_r-1.07)}} \right]
 \end{aligned}
 \tag{45}$$

These expressions provide the analytical basis for generating the operational-safety surfaces and the associated probability-density, risk-rate and confidence-rate curves presented below.

4.5. Graphical analysis of operational safety

All the curves presented in this section were generated using MATLAB. The graphical results make it possible to examine the simultaneous influence of the time between failures and the technical time to repair on the operational-safety response.

4.5.1 FPU No. 1

Figure 5 presents the operational-safety curve obtained for FPU No. 1. Figure 6 shows the corresponding probability-density curve, while Figures 7 and 8 present the risk-rate and confidence-rate curves, respectively.

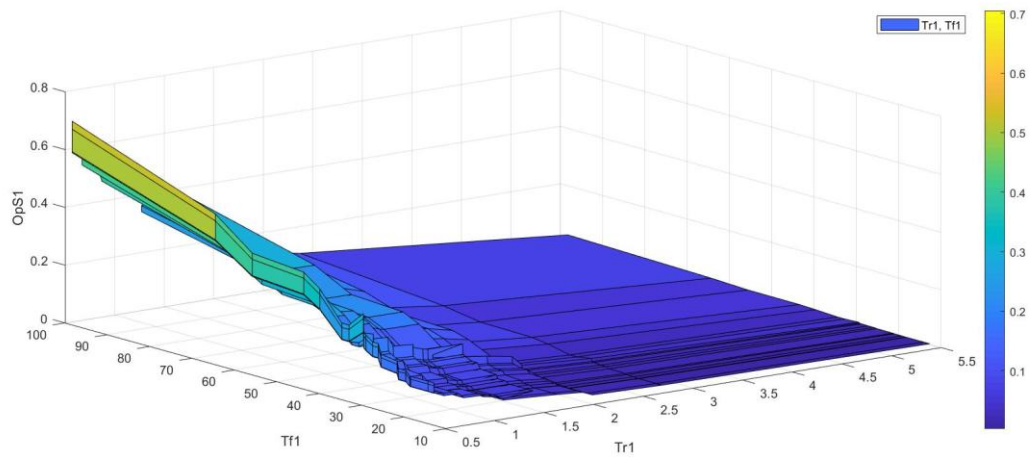


Fig. 5 Operational safety curve of FPU No. 1

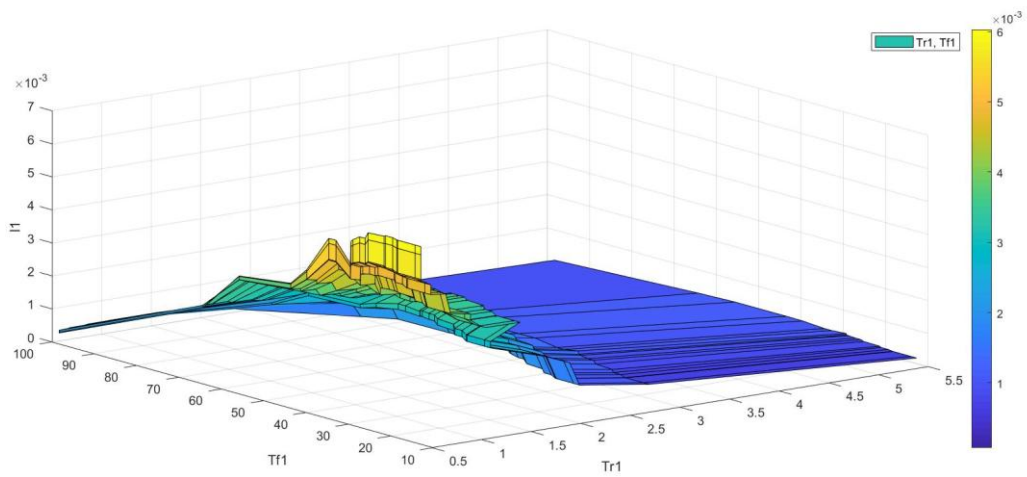


Fig. 6 Probability density curve of FPU No. 1

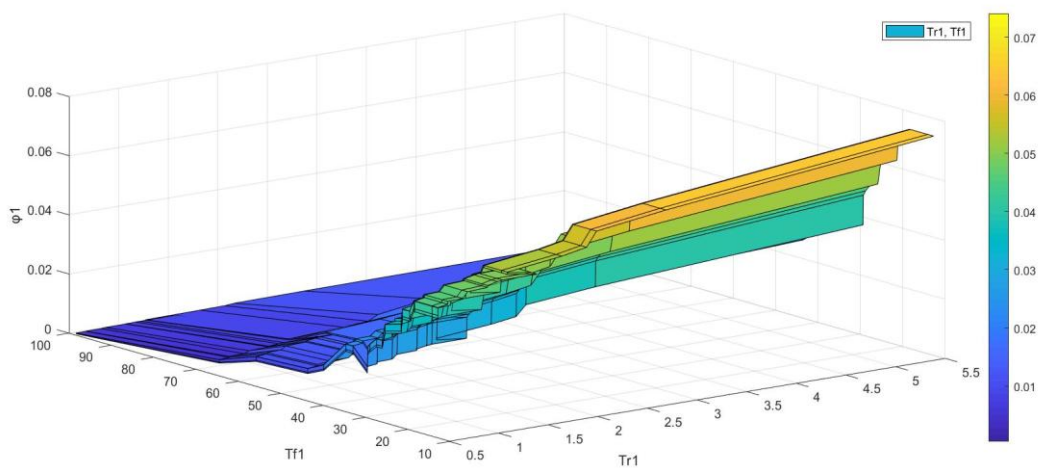


Fig. 7 Risk rate curve of FPU No. 1

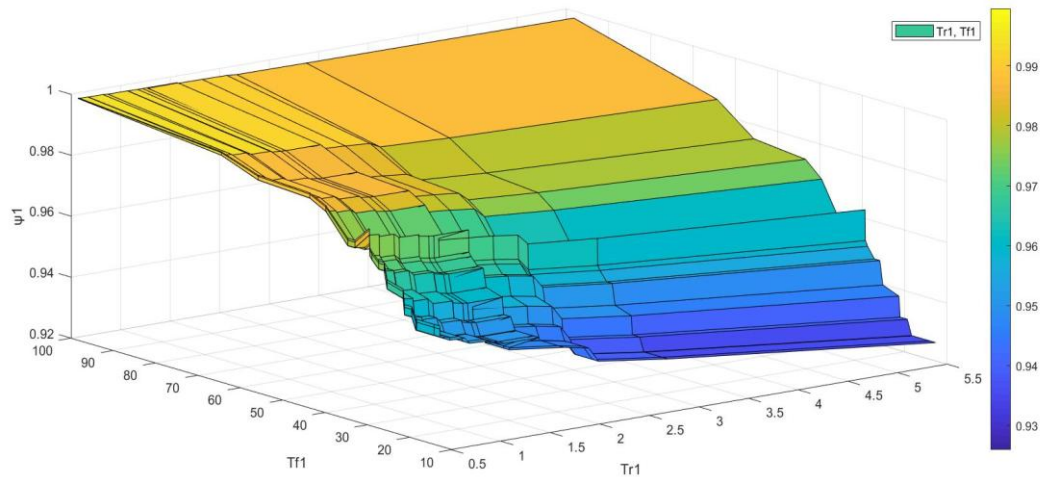


Fig. 8 Confidence rate curve of FPU No. 1

Figures 5 and 6 are obtained directly from Eqs. (40) and (41), respectively. The operational-safety response varies jointly with TBF and TTR, illustrating the combined influence of failure occurrence and repair performance on the proposed indicator.

4.5.2 FPU No. 2

Figures 9–12 present the corresponding results for FPU No. 2. Figure 9 represents operational safety, Figure 10 the probability density, Figure 11 the risk rate, and Figure 12 the confidence rate.

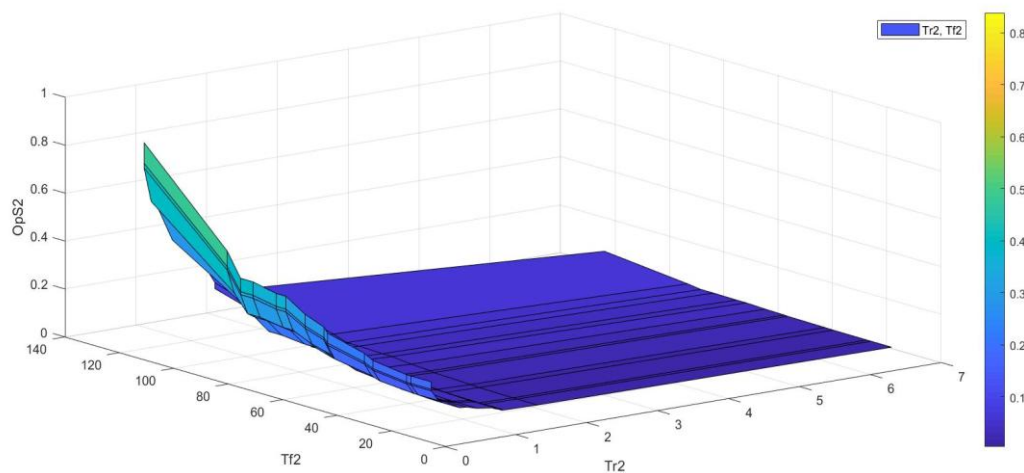


Fig. 9 Operational safety curve of FPU No. 2

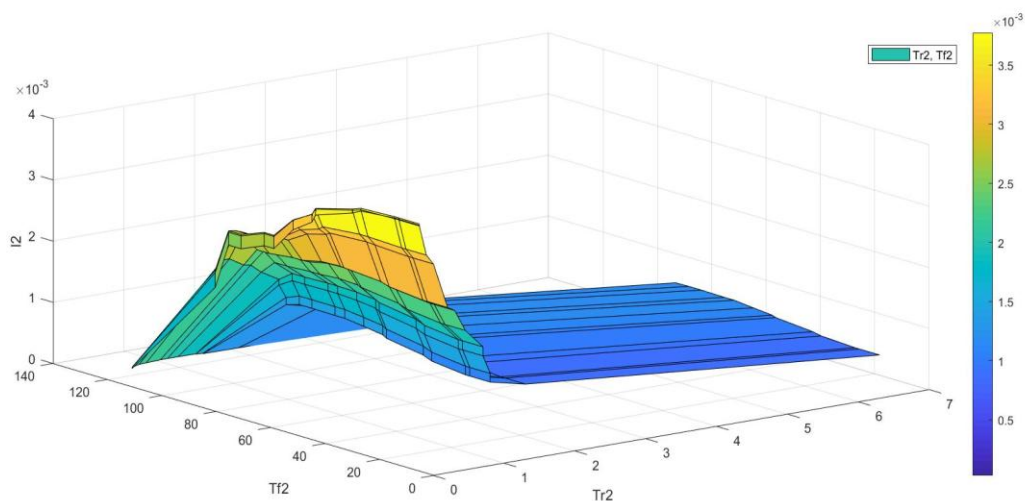


Fig. 10 Probability density curve of FPU No. 2

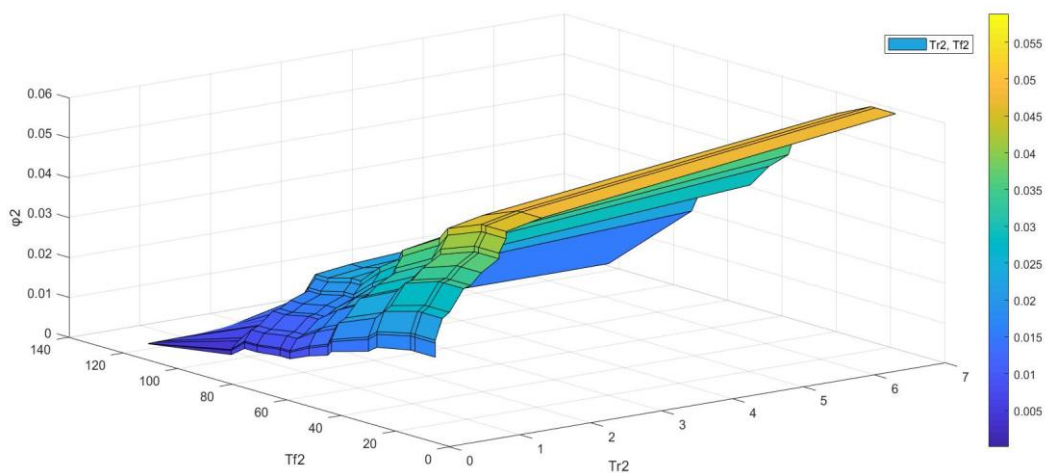


Fig. 11 Risk rate curve of FPU No. 2

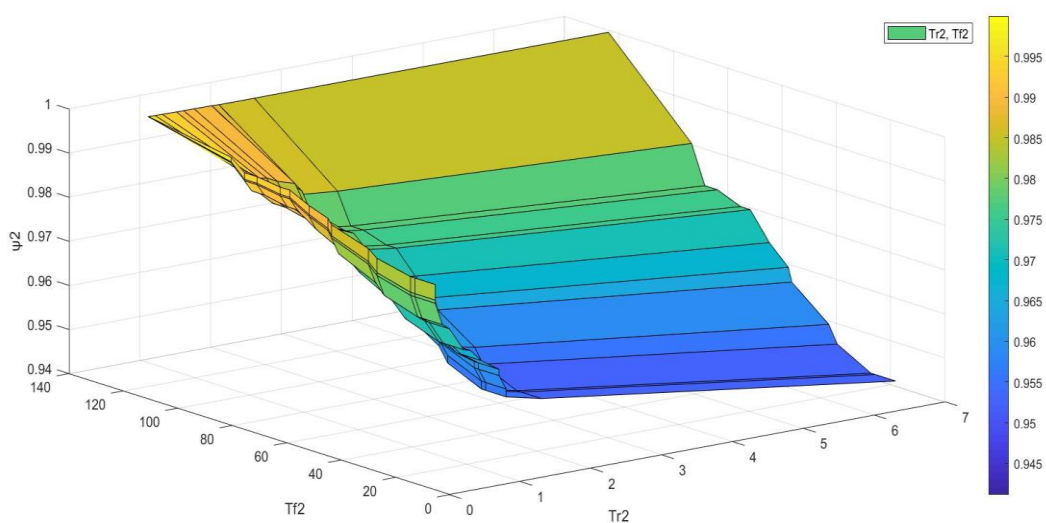


Fig. 12 Confidence rate curve of FPU No. 2

The operational-safety and probability-density curves shown in Figures 9 and 10 are derived from Eqs. (42) and (43), respectively. The graphical behavior is consistent with the probabilistic structure of the model: improved failure-time behavior contributes to higher operational safety, whereas longer repair times have an unfavorable effect on the resulting indicator.

4.5.3 FPU No. 3

Figures 13–16 show the corresponding results for FPU No. 3.

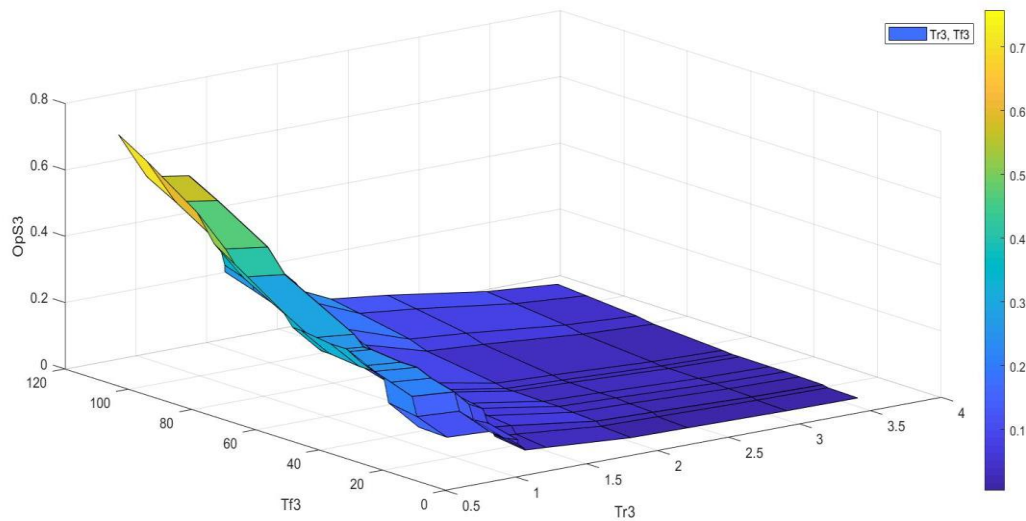


Fig. 13 Operational safety curve of FPU No. 3

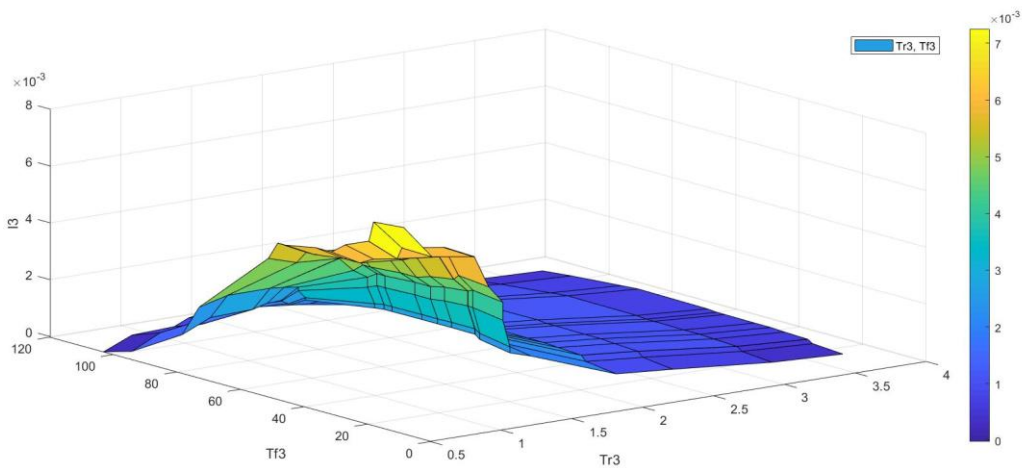


Fig. 14 Probability density curve of FPU No. 3

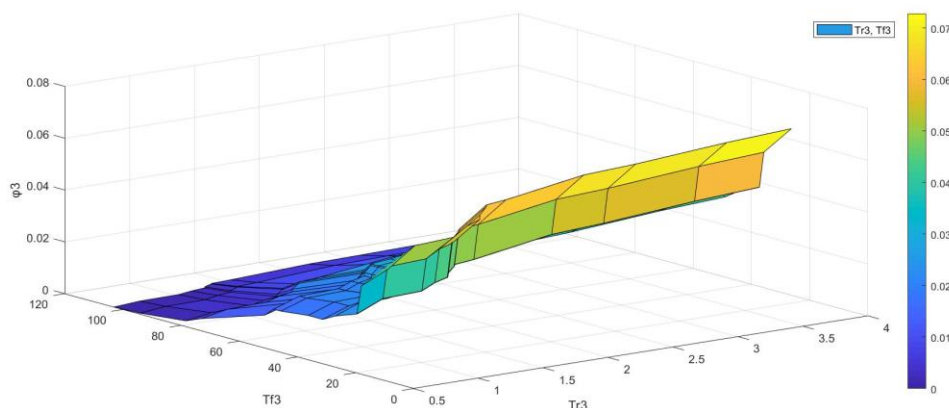


Fig. 15 Risk rate curve of FPU No. 3

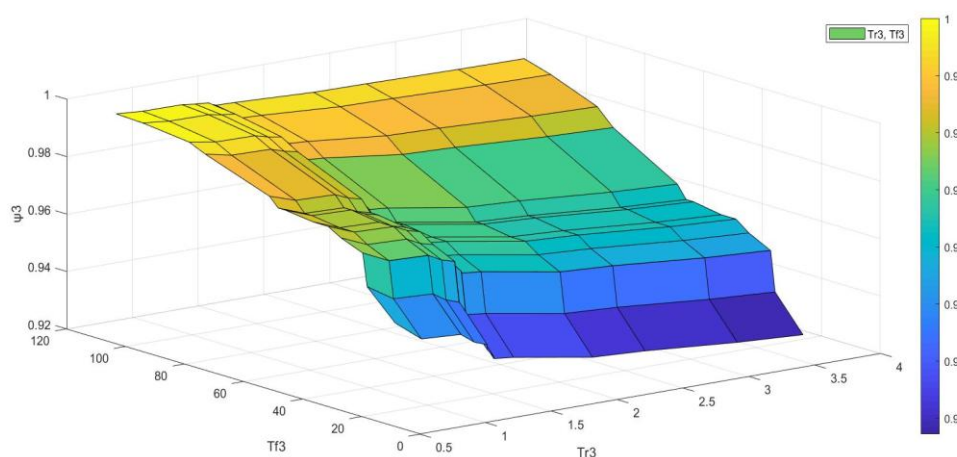


Fig. 16 Confidence rate curve of FPU No. 3

Figures 13 and 14 are derived from Eqs. (44) and (45), respectively. As for the other FPUs, the operational-safety response is strongly dependent on the combination of failure and repair times.

4.6. Comparison of the operational-safety responses

A comparison of Figures 5, 9 and 13 shows that the three FPUs exhibit a broadly similar qualitative evolution of operational safety. The corresponding probability-density curves in Figures 6, 10 and 14 also exhibit comparable overall behavior. Similar observations can be made for the risk-rate curves in Figures 7, 11 and 15 and the confidence-rate curves in Figures 8, 12 and 16. Despite this common qualitative behavior, the three FPUs do not have identical probabilistic characteristics. Their different Weibull and Gumbel parameter values modify the location and magnitude of the corresponding responses. The differences are therefore attributable to the specific failure and repair characteristics of each unit.

The operational-safety values obtained from the model remain relatively low, with maximum values reaching approximately 0.6 over the investigated range. This behavior is consistent with the more restrictive nature of the operational-safety formulation adopted in the present study, which combines the effects of reliability and maintainability through the proposed probabilistic structure.

4.7. Influence of failure and repair times

The graphical results demonstrate that operational-safety is strongly influenced by both TBF and TTR. For relatively short TBF values, the probability of maintaining the equipment in an acceptable operational condition is lower. As the TBF increases, the operational-safety indicator generally improves. This behavior is consistent with the interpretation of reliability: longer failure-free operating periods correspond to improved reliability.

The influence of TTR acts in the opposite direction. Longer repair times reduce the ability of the equipment to return rapidly to an operational state and therefore negatively affect the maintainability contribution to operational safety. The joint consideration of these two variables is one of the main characteristics of the proposed formulation. Instead of evaluating reliability and maintainability independently, the model provides a combined operational-safety indicator whose value changes according to both failure and repair behavior.

4.8. Risk-rate and confidence-rate behavior

The probability-density, risk-rate and confidence-rate results provide complementary information about the operational-safety response. The probability-density functions shown in Figures 6, 10 and 14 indicate the regions in which the operational-safety response is most concentrated. In general, regions associated with lower operational-safety values correspond to unfavorable combinations of failure and repair times. The risk-rate curves shown in Figures 7, 11 and 15 exhibit an inverse qualitative relationship with operational safety. As operational safety increases, the associated risk rate decreases over the corresponding domain. The confidence-rate curves in Figures 8, 12 and 16 remain high, with values reported in the approximate range of 0.94 – 1. These values indicate a high confidence level associated with the operational-safety estimates within the proposed probabilistic formulation. They should, however, be interpreted as a property of the estimation framework rather than as an independent experimental measure of equipment accuracy.

4.9. Comparison with the previous operational-safety formulation

An important point in interpreting the present results is the distinction between the operational-safety formulation used here and that developed in the authors' previous study. The reliability and maintainability distributions are based on the same fundamental probabilistic concepts, and the present case study uses the same historical TBF dataset. Nevertheless, the operational-safety model itself is different. The difference originates from the event formulation introduced in Eq. (9), which leads to the analytical development in Eqs. (10) – (15) and subsequently to the Weibull–Gumbel formulation of Eq. (21). Consequently, the operational-safety values obtained in the present study should not be interpreted as a simple recalculation of the previous model. They result from a distinct mathematical formulation applied to the same type of reliability and maintainability information. This distinction is particularly relevant for design applications because changing the operational-safety formulation changes the criterion used to discriminate between alternative design solutions.

4.10. Implications for mechanical design

The results demonstrate the potential of the proposed operational-safety indicator as a design-oriented criterion. At the design stage, a mechanical system can be characterized by estimated failure and repair-time distributions. The corresponding reliability and maintainability parameters can then be introduced into the proposed operational-safety model. The resulting

indicator provides an additional basis for comparing candidate solutions. In this context, a design solution should not be assessed solely according to its functional performance. Its expected failure behavior and its ability to be restored after failure should also be considered. The operational-safety criterion therefore provides a means of linking probabilistic operational-safety analysis with design evaluation. The results obtained for the three FPU's illustrate this principle using real historical operational information. They also provide the numerical basis for the design-oriented integration framework presented subsequently in Section 5.

4.11. Main findings

The application to the three FPU's leads to four principal findings. First, the K–S goodness-of-fit results support the use of the Weibull distribution for the observed TBF data at the 5% significance level. Second, the estimated Weibull shape parameters are greater than unity, indicating an increasing failure-rate regime for the three FPU's and suggesting a wear-out tendency. Third, the proposed operational-safety formulation produces relatively low operational-safety values over the investigated domain, with values reaching approximately 0.6, while the confidence-rate values remain high. Finally, the graphical results confirm that TBF and TTR jointly influence the operational-safety response. This makes the proposed indicator potentially useful for design evaluation, where both failure behavior and repair performance need to be considered.

5. Integration of operational safety into the mechanical design process

The preceding results show that the proposed operational-safety indicator can be evaluated from reliability and maintainability information obtained from operating data. The next step is to use this indicator during the mechanical design process so that operational safety is considered not only after commissioning, but also when design alternatives are being defined and assessed. The objective is therefore to establish a design-oriented framework in which operational-safety performance constitutes an additional criterion for evaluating and validating a mechanical system.

5.1. Principle of the proposed design-oriented approach

In conventional mechanical design, the selection of a design solution is generally based on functional requirements, geometric constraints, strength, manufacturability, cost and other technical criteria. Reliability and maintainability are often evaluated separately or introduced at a later stage. The approach proposed in this study integrates operational safety into this decision process by using the estimated reliability and maintainability characteristics of the system or its critical components (Ngnassi Djami et al., 2020a).

For a candidate design, the procedure consists in determining the expected failure and repair behavior, estimating the corresponding reliability and maintainability parameters, and then calculating the operational-safety indicator according to the formulation established in Section 2. The resulting value can subsequently be compared with the operational-safety requirement associated with the intended application.

The general principle can be represented as:

Candidate mechanical design → Reliability assessment → Maintainability assessment
→ Operational-safety calculation → Comparison with requirement → Design decision

Thus, operational safety becomes a design-evaluation criterion rather than an indicator considered exclusively during operation.

5.2. Proposed procedure

The proposed integration procedure comprises the following main stages.

Step 1 — Definition of design requirements.

The functional, operational, environmental and safety requirements of the mechanical system are first established. Particular attention is given to the required level of operational safety, denoted by $MOpS_{req}$, when such a requirement is specified.

Step 2 — Collection of reliability and maintainability information.

The available failure and repair information is collected for the system or for comparable equipment. The principal variables considered are the time between failures (TBF) and the time to repair (TTR). When sufficient historical data are available, they are used directly for statistical estimation. In the absence of sufficient operational data, simulation or reliability information from analogous systems can be considered.

Step 3 — Extraction of design information.

The mechanical design is examined using the available design documentation, including CAD models, technical drawings, assemblies and component specifications. This step makes it possible to identify the elements whose failure or repair characteristics may significantly influence the operational behavior of the system.

Step 4 — Identification of critical components.

The components that have a significant influence on system reliability, maintainability or operational safety are identified. Particular attention is given to components associated with frequent failures, long repair times or severe operational consequences.

Step 5 — Selection of probability distributions.

Appropriate probability laws are selected to represent the observed failure and repair behavior. In the present study, the Weibull distribution is used for failure behavior and the Gumbel distribution for repair behavior. The choice of distributions should nevertheless be supported by statistical goodness-of-fit analysis rather than imposed independently of the data.

Step 6 — Estimation of model parameters.

The parameters of the selected distributions are estimated from the available observations. For the reliability model, the Weibull parameters are estimated using the maximum likelihood method. The maintainability parameters are determined from the repair-time data using the corresponding Gumbel representation.

Step 7 — Calculation of operational-safety indicators.

The estimated parameters are introduced into the operational-safety formulation developed in this study. The resulting operational-safety function can then be evaluated for relevant operating and repair-time conditions. Additional indicators, such as the probability density, risk rate and

confidence rate, can also be examined to obtain a more complete characterization of system behavior.

Step 8 — Decision on the design solution.

The calculated operational-safety level is compared with the design requirement. A candidate solution satisfying the specified requirement can be retained. If the requirement is not achieved, the design must be reconsidered.

Step 9 — Design modification.

When the operational-safety requirement is not satisfied, modifications can be introduced at the design level. These modifications may concern component selection, architecture, accessibility for maintenance, redundancy, dimensions, material selection or other design characteristics affecting failure and repair behavior.

Step 10 — Reassessment and validation.

The modified design is reassessed using the same probabilistic framework. The process is repeated until an acceptable solution is obtained. The final design is then subjected to the appropriate technical and functional validation procedures.

5.3. Integration of the operational-safety indicator into design decisions

The proposed framework considers operational safety as a performance measure associated with a candidate design. Let MOS_{syst} denote the mean operational-safety measure evaluated for the system. The design decision can consequently be expressed conceptually as

$$MOpS_{\text{syst}} \geq MOpS_{\text{req}}$$

when a minimum operational-safety requirement is defined.

This criterion does not replace conventional mechanical design constraints. Rather, it complements them. A design solution should simultaneously satisfy its functional, structural, manufacturing, economic and operational requirements. The operational-safety indicator therefore provides an additional quantitative basis for comparing alternative solutions.

For example, when two candidate designs provide equivalent functional performance, the design exhibiting the higher operational-safety level under comparable operating conditions may be preferred. Conversely, a design presenting a lower operational-safety level can be subjected to targeted modifications before its physical implementation.

5.4. Role of simulation when operational data are insufficient

A major difficulty in applying probabilistic safety criteria during the early stages of design is the limited availability of operational data. A new mechanical system may not yet have accumulated sufficient failure and repair observations to permit reliable parameter estimation.

In such circumstances, the proposed framework can be supplemented by Monte Carlo simulation. Probabilistic distributions representative of the expected failure and repair behavior can be assigned to the relevant components, and repeated simulations can be performed to estimate the corresponding operational-safety response.

This approach allows the designer to investigate the influence of uncertain failure and repair characteristics before sufficient field data become available. As operational data are progressively accumulated, the assumed distributions and their parameters can subsequently be replaced or updated using observed TBF and TTR data.

5.5. Design-oriented metamodel

The overall integration of the proposed approach into mechanical design is summarized by the design-oriented metamodel presented in Fig. 17.

Legend :

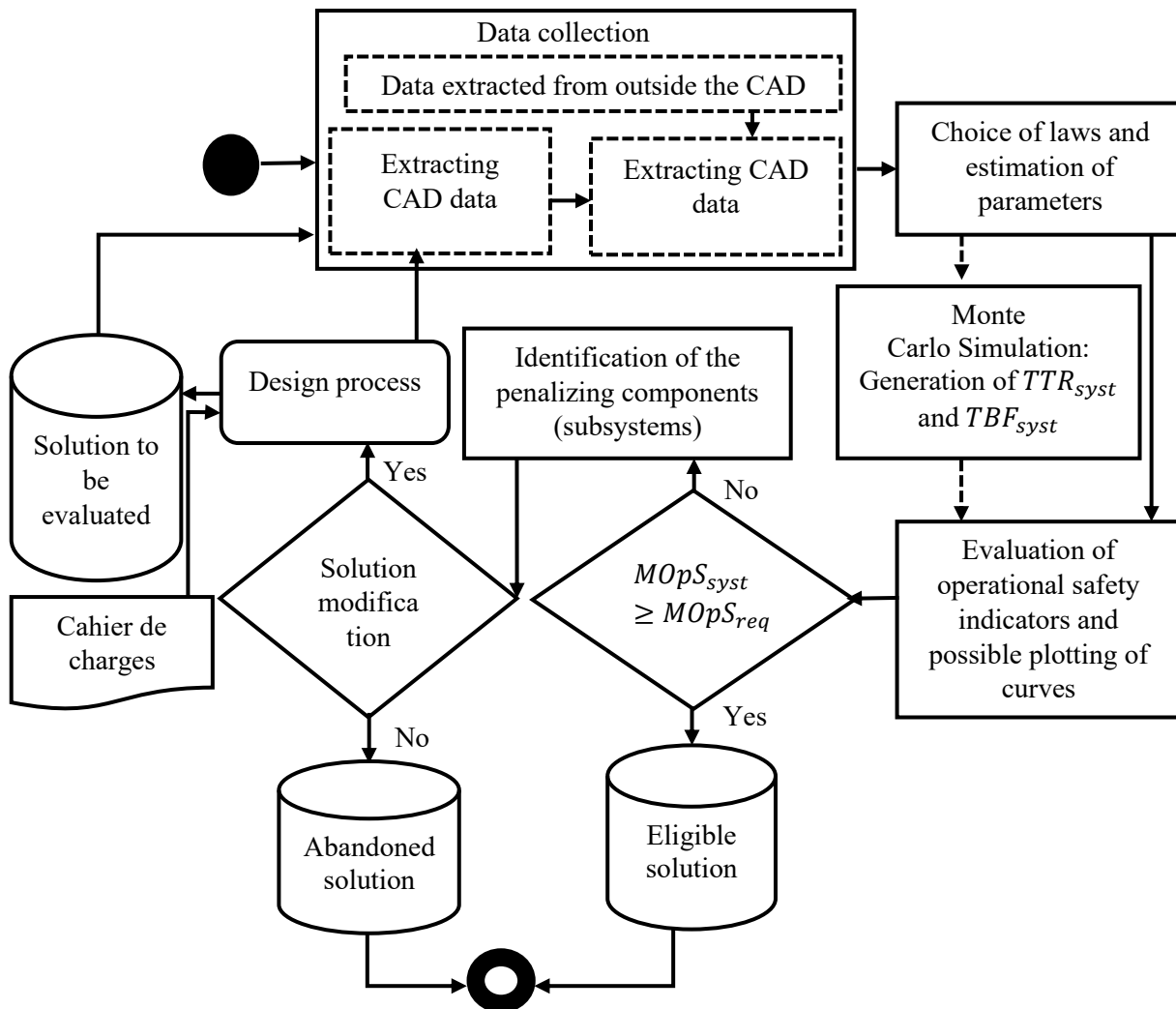
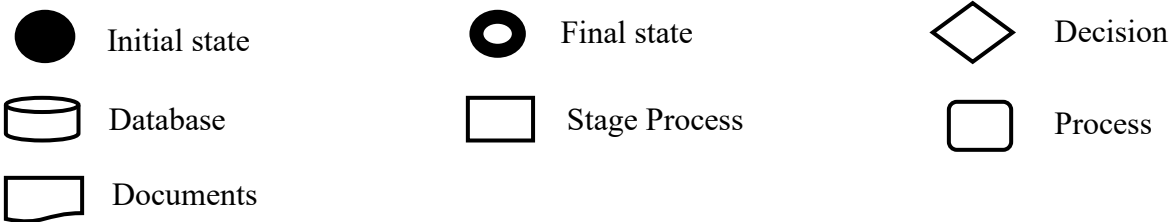


Fig. 17. Design-oriented framework for integrating operational safety into the mechanical design process.

The metamodel establishes a feedback loop between design definition, probabilistic assessment and design modification. The design is initially characterized through its mechanical and functional parameters. Reliability and maintainability information is then used to evaluate operational safety. The resulting assessment provides feedback to the design stage when the required performance is not achieved.

This feedback mechanism is particularly relevant for mechanical systems in which failures and repair operations can significantly affect service continuity and operational performance. It also provides a structured way of incorporating field experience into subsequent design iterations.

5.6. Contribution and practical significance

The principal contribution of this framework is to establish a direct link between probabilistic operational-safety assessment and mechanical design decision-making. Unlike an assessment performed solely after equipment commissioning, the proposed approach provides a mechanism for using reliability and maintainability information to support design evaluation and subsequent modification.

The approach is particularly suitable for mechanical systems for which failure and repair data are available from existing equipment or analogous installations. In the context of the present case study, the historical data from the fire pump units provide the statistical basis for demonstrating how the proposed operational-safety formulation can be evaluated and subsequently incorporated into a design-oriented process.

However, the framework presented here should be regarded as a design-oriented evaluation and validation methodology, rather than as a complete experimental validation of alternative mechanical designs. Its application to several competing designs, followed by physical testing and long-term operational monitoring, would constitute an important extension of the present work.

6. Conclusion

This study proposed a probabilistic and design-oriented approach for evaluating and integrating operational safety into the mechanical design process. The methodology is based on the joint consideration of reliability and maintainability, respectively described by Weibull and Gumbel distributions, with the parameters of the reliability model estimated using the maximum likelihood method. The resulting operational-safety formulation provides a quantitative indicator that can be evaluated from failure and repair data.

The application to the three fire pump units of the CCPD demonstrated the applicability of the proposed approach. The historical TBF and TTR data allowed the reliability and maintainability characteristics of the equipment to be characterized and subsequently used to calculate the operational-safety functions, probability densities, risk rates and confidence rates. The results obtained showed differentiated operational-safety behavior among the three units, while the statistical analyses supported the suitability of the adopted probabilistic models.

An important aspect of this work is that the operational-safety formulation proposed here is different from that developed in our previous study. Although both approaches use reliability and maintainability as fundamental probabilistic quantities and rely on the same general estimation strategy, the difference in the formulation of the operational-safety model leads to

different analytical developments and assessment results. This distinction constitutes an essential element of the present contribution.

The proposed approach nevertheless has some limitations. Its application has been illustrated using historical operational data from existing equipment, and the complete validation of the methodology on alternative candidate mechanical designs has not yet been performed experimentally. Future work should therefore focus on applying the framework to several competing designs, incorporating additional sources of uncertainty, and validating the predicted operational-safety performance through experimental testing and long-term field data.

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Data availability: The authors do not have permission to share data.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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